ANSWER KEY & MARKING SCHEME SET A

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a and b are not both zero	definition of standard form
2	**(d)** 0	horizontal line; rise is zero
3	**(a)** Both A and R are true and R is the correct explanation of A	R is the reason A holds
4	**(b)** Both A and R are true but R is not the correct explanation of A	both true, but the slope does not explain the intercept
5	**(c)** 7	solve the pair first
6	**(c)** -1	rise ÷ run
7	**(c)** 1/2	write as y = mx + d, so m = −a/b
8	**(c)** (2, 3)	substitute each point in the equation
9	**(a)** a unique solution	compare a1/a2, b1/b2, c1/c2
10	**(a)** 2	3 + 2k = 7

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5x - 3y - 7 = 0$ (1). So $a = 5$, $b = -3$, $c = -7$ (or all signs reversed) (1).	1+1
12	Substituting: $4(3) + 5(-2) = 12 - 10 = 2$ (1). The equation is satisfied, so $(3, -2)$ is a solution (1).	1+1
13	Put $x = 0$: $-2y = 14$, $y = -7$, so $(0, -7)$ (1). Put $y = 0$: $7x = 14$, $x = 2$, so $(2, 0)$ (1).	1+1
14	Substitute: $3a(2) + 2(-1) = 4$ (1). $6a - 2 = 4$, so $a = 1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	From the first equation $y = 11 - 2x$ (1). Substituting in the second: $x - 3(11 - 2x) = -5$, so $7x = 28$ and $x = 4$ (1). Then $y = 11 - 8 = 3$; check: $4 - 9 = -5$. Solution $(4, 3)$ (1).	1+1+1
16	Multiply the second by 2: $10x - 4y = 8$ (1). Add to the first: $13x = 26$, so $x = 2$ (1). Then $4y = 18 - 6 = 12$, $y = 3$; check $10 - 6 = 4$ (1).	1+1+1
17	Let apples cost ₹x and grapes ₹y per kg: $3x + 2y = 540$, $2x + 5y = 690$ (1). Eliminating x: $(\times 2)\ 6x + 4y = 1080$, $(\times 3)\ 6x + 15y = 2070$, so $11y = 990$, $y = 90$ (1). Then $3x = 540 - 180 = 360$, $x = 120$. Apples ₹120/kg, grapes ₹90/kg (1).	1+1+1
	OR: $x + 8y = 140$ and $x + 12y = 200$ (1). Subtracting: $4y = 60$, $y = 15$ and $x = 140 - 120 = 20$ (1). Fare for 20 km $= 20 + 20 \times 15 =$ ₹320 (1).	1+1+1
18	Let the digits be x (tens), y (units). $x + y = 11$ (1). $(10x + y) - (10y + x) = 27$, so $x - y = 3$ (1). Then $x = 7$, $y = 4$ and the number is 74 (1).	1+1+1
	OR: Let the son be s and the father f years: $f = 4s$ (1). $f + 5 = 3(s + 5)$, so $f - 3s = 10$ (1). Substituting, $4s - 3s = 10$, $s = 10$ and $f = 40$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Multiply the first by 3 and the second by 5: $9x - 15y = -3$ and $10x + 15y = 60$ (2). Add: $19x = 57$, so $x = 3$ (1). Then $3y = 12 - 6 = 6$, $y = 2$ (1). Verification: $9 - 10 = -1$ and $6 + 6 = 12$ (1).	2+1+1+1
	OR: For $x + 2y = 8$: $(0, 4)$, $(8, 0)$, $(2, 3)$ (1). For $2x - y = 1$: $(0, -1)$, $(1, 1)$, $(2, 3)$ (1). Plotting and drawing the two lines (1), they meet at $(2, 3)$ so $x = 2$, $y = 3$ (1). The lines intersect at exactly one point, so the solution is unique (1).	1+1+1+1+1
20	No solution needs $\frac{k}{16} = \frac{4}{k} \neq \frac{k-4}{k}$ (1). $k^2 = 64$, so $k = \pm 8$ (1). For $k = 8$: $\frac{k-4}{k} = \frac{1}{2} = \frac{8}{16}$, so all three ratios are equal and the lines coincide (1). For $k = -8$: $\frac{k}{16} = -\frac{1}{2}$ but $\frac{k-4}{k} = \frac{-12}{-8} = \frac{3}{2}$, so the ratios of the constants differ (1). Hence $k = -8$ and the graphs are parallel lines (1).	1+1+1+1+1

	OR: (i) First: slope $\frac{1}{2}$, y-intercept 2; second: slope -1, y-intercept 5 (1). (ii) $\frac{1}{2}x + 2 = -x + 5$, so $\frac{3}{2}x = 3$ and $x = 2$ (1). $y = 3$, so the lines meet at $(2, 3)$ (1). Check $\frac{1}{2}(2) + 2 = 3$ and $-2 + 5 = 3$ (1). (iii) The slopes differ, so the lines intersect in exactly one point and the pair has a unique solution (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET B

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** infinite	any value of x gives a value of y
2	**(a)** parallel lines	no common point
3	**(a)** Both A and R are true and R is the correct explanation of A	ratios 1/2 = 1/2 ≠ 5/11
4	**(d)** (3, 2)	substitute the pair in both equations
5	**(d)** 5	solve the pair first
6	**(d)** 2	rise ÷ run
7	**(d)** 4	put x = 0
8	**(b)** (1, 2)	substitute each point in the equation
9	**(d)** infinitely many solutions	compare a1/a2, b1/b2, c1/c2
10	**(b)** 19	x + y = 30, x − y = 8 give x = 19

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-5}{4-(-2)}$ (1) $= \frac{-6}{6} = -1$ (1).	1+1
12	$2y = -4x + 10$, so $y = -2x + 5$ (1). Slope $= -2$ and y-intercept $= 5$ (1).	1+1
13	(i) $y = -3$ is horizontal; rise is 0, so slope $= 0$ (1). (ii) $x = 2$ is vertical; run is 0, so the slope is undefined (1).	1+1
14	Substituting $x = 0,\ y = 0$ gives $0 - 0 = 0$ (1). So $(0, 0)$ satisfies the equation and lies on the line; this happens whenever $c = 0$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $\frac{2}{4} = \frac{-3}{-6} = \frac{5}{10}$: infinitely many solutions (1). (ii) $\frac{3}{6} = \frac{2}{4} = \frac{1}{2} \neq \frac{-7}{-9}$: no solution (1). (iii) $\frac{1}{2} \neq \frac{2}{-1}$: a unique solution (1).	1+1+1
16	For infinitely many solutions $\frac{k}{3} = \frac{2}{6} = \frac{4}{12}$ (1). $\frac{2}{6} = \frac{1}{3}$, so $k = 1$ (1). Then the second equation is 3 times the first, so the lines coincide and there is no single intersection point (1).	1+1+1
17	Slope $= \frac{7-3}{3-1} = 2$ (1). Using $(1, 3)$: $3 = 2(1) + d$, so $d = 1$ (1). Equation: $y = 2x + 1$; check $(3, 7)$: $2(3) + 1 = 7$ (1).	1+1+1
	OR: $x + y = 6$: points $(0, 6)$ and $(6, 0)$; $x - y = 2$: points $(0, -2)$ and $(2, 0)$ (1). Drawing the two lines, they cross at $(4, 2)$ (1). A unique solution $x = 4$, $y = 2$, since the lines intersect at one point (1).	1+1+1
18	$l + w = 30$ and $l - w = 6$ (1). Adding: $2l = 36$, $l = 18$ m (1). Then $w = 12$ m; check: $18 - 12 = 6$ (1).	1+1+1
	OR: A: $60 + 2t$; B: $20 + 4t$ (1). Equal when $60 + 2t = 20 + 4t$, so $t = 20$ minutes (cost ₹100) (1). For $t > 20$ minutes Plan A is cheaper; for $t < 20$ Plan B is cheaper (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Incomes $5x, 4x$; expenditures $3y, 2y$ (1). $5x - 3y = 3000$ and $4x - 2y = 3000$ (1). Multiply by 2 and 3: $10x - 6y = 6000$, $12x - 6y = 9000$; subtract: $2x = 3000$, $x = 1500$ (1). Then $3y = 7500 - 3000 = 4500$, $y = 1500$ (1). Incomes ₹7500 and ₹6000; expenditures ₹4500 and ₹3000 (1).	1+1+1+1+1
	OR: $5x + 8y = 9300$ and $3x + 6y = 6000$ (2). Simplify the second: $x + 2y = 2000$, so $x = 2000 - 2y$ (1). Substituting: $10000 - 10y + 8y = 9300$, $2y = 700$, $y = 350$ (1). Then $x = 1300$: a bat costs ₹1300 and a ball ₹350 (1).	2+1+1+1
20	Let the fixed charge be ₹x and the extra charge ₹y per day: $x + 4y = 36$ and $x + y = 24$ (2). Subtracting: $3y = 12$, $y = 4$ (1). Then $x = 20$ (1). For 10 days: $20 + 7 \times 4 =$ ₹48 (1).	2+1+1+1
	OR: (i) P: $20 + 5g$; Q: $5 + 10g$ (1). (ii) $20 + 5g = 5 + 10g$, so $5g = 15$ and $g = 3$, with cost ₹35 (1). (iii) For $g = 2$: P costs ₹30 and Q costs ₹25, so Q is cheaper (1). (iv) Line P has slope 5 and intercept 20; line Q has slope 10 and intercept 5 (1). The steeper line Q starts lower but rises faster, so they cross at $(3, 35)$ (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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